

# Predicting Employee Turnover at Salifort Motors

Executive Summary · HR Retention Analysis

Salifort Motors loses about 1 in 6 employees, and replacing them is costly. Survey data from ~12,000 employees was analysed to predict who is likely to leave and why. Four models were built and tuned; a **tuned Random Forest** was selected as the champion.

PROJECT STATUS — CHAMPION MODEL: TUNED RANDOM FOREST

0.88

Recall

Catches ~88% of leavers

0.91

Precision

Right ~91% when flagging someone

0.90

F1 Score

Balanced precision/recall

0.97

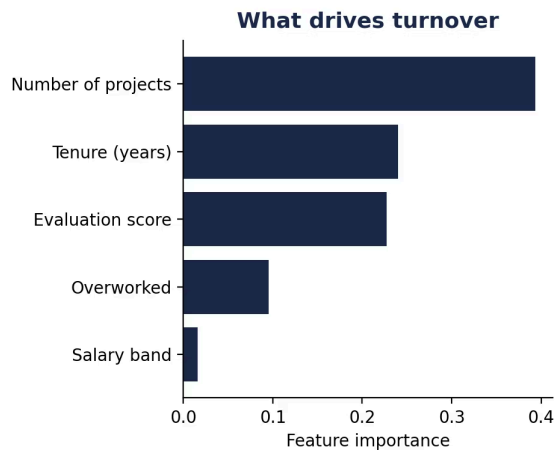
AUC

Near-perfect discrimination

✓ **Benefit:** Ranks the factors behind every prediction — HR gets a reason, not just a flag.

⚠ **Limitation:** Learns from past data; must be re-checked periodically. Excludes self-reported satisfaction to stay reliable when that survey is missing.

## KEY INSIGHTS



Random Forest feature importance — drivers of turnover

### Workload

Projects + monthly hours is the strongest predictor

### Tenure

Risk peaks sharply around year 5

### Performance

High review scores correlate with departure risk

### Overwork

175+ hours/month shows elevated risk

### Pay Gap

Low-paid staff leave ~4x more often

### Three Clusters

Burnout, under-used (2 projects), overloaded high performers

## NEXT STEPS

01

### Cap Workloads

Review anyone on 6–7 projects or 240+ hours/month

02

### Re-Engage the Under-Used

Assign meaningful work to low-satisfaction group

03

### Act at the 4–5 Year Mark

Schedule pay, promotion, and growth reviews proactively

04

### Decouple Recognition from Hours

Reward impact, not volume

05

### Deploy Model Responsibly

Supportive outreach only, with ongoing monitoring

All interventions should be implemented with transparency. The model is a tool for support — not surveillance.